Sébastien Naze, PhD

COMPUTATIONAL NEUROSCIENTIST

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Manhattan, New York

French citizen
U.S. Green card holder

"The best way to make your dreams come true is to wake up."

Paul Valéry

PROGRAMMING SKILLS

Python, C/C++, Java
Distributed and parallel computing (MPI, DASK)
Multi-Objective Optimization (MCMC, SMC)
Object Oriented & Service Architectures
Cloud Computing (docker, AWS, SQLite, Redis)
Tensor computing (PyTorch, Einops)
Agentic Coding – Mech Interp – Evals
Jupyter Lab / Data Eng. (Pandas, Marimo, …)
Agile methodology – CI/CD – Git

SCIENTIFIC SKILLS

Time series analysis (Fourier, Wavelets)
Machine Learning (classical, LLMs, RL, EBMs)
Network Science (Graph Signal Processing)
Dynamical Systems Modeling & Analysis
Neuroimaging & Bayesian Methods

COMMUNICATION & LEADERSHIP

Client presentations & Tutorials
Student / Intern Supervision
Fluent in English, French & Spanish

PROFESSIONAL EXPERIENCE

QIMR RESEARCH FELLOW

Queensland Institute of Medical Research, Brisbane, Australia *2021 - current*

- Data engineering, analysis of multimodal dataset, multiscale modeling, and funding acquisition for neurostimulation in obsessive-compulsive disorders.
- Development of Neuro-AI pipeline for clinical prediction and mechanistic interpretability.

IBM RESEARCH SCIENTIST

IBM Research, Melbourne, Australia *2019 - 2021*

- Designed and implemented an interface between brain modeling and machine learning algorithms in Python and C++.
- Performed parameter optimization using evolutionary algorithms.
- Integrated libraries into service architecture for cloud computing.
- Worked in international team of 4-12 people operating from multiple countries.

IBM POSTDOCTORAL RESEARCHER

IBM T. J. Watson Research Center, New York, USA *2016 – 2019*

- Delivered IBM Solutions in brain modeling to clients (CHDI, Pfizer) investigating Huntington's disease phenotypes in-silico.
- Delivered quarterly reports, client presentations and software patches.
- Served as team co-lead on collaborative research projects with academic partners (Stanford, Brown University, Univ. of Oxford, UHN Toronto).

RESEARCH ASSISTANT

Institute of Systems Neuroscience, Marseille, France *2011 – 2015*

Preparation of PhD Thesis entitled Multiscale Modeling of Epileptic Seizure Dynamics.

TECHNICAL ASSISTANT

Vrije Universiteit, Amsterdam, The Netherlands *2010 – 2011*

Preparation of MSc Thesis entitled Computational Agent Modeling of Post-Traumatic Stress Disorders.

EDUCATION

PHD COMPUTATIONAL NEUROSCIENCE

Aix-Marseille Université, France
2015

MSC INFORMATION SCIENCE

Vrije Universiteit, The Netherlands
2011

BSC SOFTWARE ENGINEERING

Heriot-Watt University, United Kingdom
2009

PUBLICATIONS



Naze S, et al. (2025) Mechanisms and interventions promoting healthy frontostriatal dynamics in obsessive-compulsive disorder. *Nature Communications.*

Webb L., *(…),* **Naze S**., Cocchi L. (2025) Altered striatal functional gradients in obsessive-compulsive disorder. *Biological Psychiatry.*

L. Cocchi, **Naze S**, et al. (2023) Effects of noninvasive stimulation of the rostromedial prefrontal cortex on symptoms and frontostriatal connectivity in obsessive-compulsive disorder. *Nature Mental Health*.

Naze S, et al. (2023) Mechanisms of imbalanced frontostriatal functional connectivity in obsessive-compulsive disorder. *Brain.*

Sihag S, **Naze S**, et al. (2022) Functional brain activity constrained by structural connectivity reveals cohort-specific features for serum neurofilament light chain. *Nature Communications Medicine*.

Naze S, et al. (2021) Features importance in seizure classification using scalp EEG reduced to single timeseries. *IEEE EMBC.*

Naze S, et al. (2021) Robustness of connectome harmonics to local gray matter and long-range white matter connectivity changes. *Neuroimage*.

Sihag S, **Naze S**, et al. (2020) Multimodal Dynamic Brain Connectivity Analysis Based on Graph Signal Processing for Former Athletes With History of Multiple Concussions. *IEEE Transactions on Signal and Information Processing over Networks.*

Sihag S, **Naze S**, et al. (2019) GSP Analysis of Brain Imaging Data from Athletes with History of Multiple Concussions. *IEEE ISIP.*

Naze S, et al. (2019) Classification of TMS evoked potentials using ERP time signatures and SVM versus deep learning. *IEEE EMBC.*

Naze S, et al. (2018) Cortico-striatal cross-frequency coupling and gamma genesis disruptions in Huntington's disease mouse and computational models. *eNeuro 5:6*.

Naze S, et al. (2015) Computational modeling of seizure dynamics using coupled neuronal networks: factors shaping epileptiform activity. PLoS Comp Biol 11:5.

Bernard C, **Naze S**, et al. (2014) Modern concepts of seizure modeling. *Annual Reviews on Neurobiology* 114:120-153.

Naze S, Treur J. (2012) A Computational Agent Model for Post-Traumatic Stress Disorders. *International Conference on Neural Information Processing*.

Naze S, Treur J. (2011) A computational model for development of post-traumatic stress disorders by hebbian learning. *Biologically-inspired Cognitive Architectures.*

CONFERENCE PRESENTATIONS

Naze S. (2025) Frontostriatal dynamics modelling obsessions and compulsions. *Cognitive Computational Neuroscience (CCN),* Amsterdam, The Netherlands.

Naze S. (2025) Mechanisms and interventions promoting healthy frontostriatal dynamics in obsessive-compulsive disorder. *Organisation for Human Brain Mapping (OHBM),* Brisbane, Australia.

Naze S. (2022) Mechanisms of imbalanced fronto-striatal functional connectivity in obsessive-compulsive disorder. *Biological Psychiatry Australia,* Newcastle, Australia.

Naze S. (2021). Features importance in seizure classification using scalp EEG reduced to single timeseries. *IEEE Engineering in Medicine and Biology Conference* (Online).

Naze S. (2020). Functional requirements of intentional control over the integrated cortico-thalamo-cortical and basal ganglia systems using neural computations. *Organization for Computational Neurosciences 29th Annual Meeting* (Online).

Naze S. (2019) Classification of TMS evoked potentials using ERP time signatures and SVM versus deep learning. *IEEE Engineering in Medicine and Biology Conference,* Berlin, Germany.

Naze S. (2019) Robustness of connectome harmonics to local gray matter and long-range white matter connectivity changes. *Organization for Computational Neurosciences 28th Annual Meeting,* Barcelona, Spain.

Naze S. (2018) Sensitivity analysis of the connectome harmonics framework and applications to neurodegenerative diseases. *Organization for Computational Neurosciences 27th Annual Meeting,* Seattle, United Sates.

Naze S. (2017) Cortico-striatal cross-frequency coupling and gamma genesis disruptions in Huntington's disease mouse and computational models. *Society for Neuroscience,* Washington D.C., United States.

Naze S. (2015) Computational modeling of initiation and propagation of epileptic seizures. *7th International Workshop on Seizure Prediction,* Melbourne, Australia.

Naze S. (2011) A Computational Agent Model of Post-Traumatic Disorders. *Biologically-Inspired Cognitive Architectures,* Washington D.C., United States.

AWARDS AND GRANTS

2023 National Health and Medical Research Council of Australia (NHMRC) IDEAS Grant number 2027597:
"Noninvasive ultrasonic deep brain stimulation as a novel tool to change brain activity in obsessive-compulsive disorder"
as Chief Investigator (CIC), responsible for predictive modeling and neuroimaging analysis. ***1.246.630 $AU***

REFERENCES

Dr. James Kozloski (Manager)
IBM Research, 1101 Kitchawan Road, Yorktown Height, New York 10598, United States.
kozloski@us.ibm.com

Dr. Luca Cocchi (Principal Investigator)
QIMR Berghofer Medical Research Institute, 300 Herston Road, Brisbane 4006, Australia.
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Dr. Stefan Harrer (Chief Innovation Officer)
Digital Health CRC, 520 Bourke Street, Melbourne 3000, Australia.
stefan.harrer@dhcrc.com

Dr. Viktor Jirsa (Director)
Institut de Neurosciences des Systèmes, 27 Boulevard Jean Moulin, 13005 Marseille, France.
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Dr. Christophe Bernard (Team head)
Institut de Neurosciences des Systèmes, 27 Boulevard Jean Moulin, 13005 Marseille, France.
christophe.bernard@univ-amu.fr